ON AN INTEGRAL SCHEME A PERFECT COMPLEX BECOMES LOCALLY FREE WITH CONSTANT RANKS ON AN OPEN, AND OF TOR DIMENSION AT MOST ONE AFTER AN ADMISSIBLE BLOWUP

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project own definitions of the reference, slogan and history display environments (verbatim package plus three newenvironment lines) that the retained file carries at its end and that the project itself typesets as comments. No macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The author opens the delivered proof with a rhetorical appeal to the reader; the sentence is kept verbatim in the text (source lines 13116-13117): "We strongly urge the reader to find their own proof of the existence of U." The argument itself is then supplied in full inside the counted span.

Editorial note. Canonical location of the counted proposition: Stacks tag 0ESR (sources/stacks-flat/pinned-original-source.tex line 13106).

Allowed standard prerequisites (editorial).

- Descent of relatively perfect objects: isomorphic after a point implies isomorphic on a neighbourhood. Derived Categories of Schemes, Lemma labelled perfect-lemma-descend-relatively-perfect (Stacks tag 0DI8), invoked at proof lines 13124-13126. Supplies the open neighbourhood U on which the model complex E' and E agree; the construction of E' from the dimensions at eta is inside the counted span.
- The admissible blowup making the cohomology sheaves of a perfect complex of tor dimension at most one. Same chapter, Lemma labelled lemma-blowup-complex (source line 13036), invoked at proof lines 13127-13128. Supplies the blowup once a nonempty open with locally free constant rank cohomology is available; the reduction to it is in the counted span.

Source reference. Original lemma 0DI8: sources/stacks-chow/tags.txt, source lines 16933–16933. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 13125 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0ESQ: sources/stacks-flat/pinned-original-source.tex, source lines 13036–13036. Same-chapter pointer into the author environment lemma

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at source line 13036; invoked at proof line 13128 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let X be an integral scheme. Let $E \in D(\mathcal{O}_X)$ be perfect. Then there exists a nonempty open $U \subset X$ such that $H^i(E|_U)$ is finite locally free of constant rank r_i for all $i \in \mathbf{Z}$ and there exists a U-admissible blowup $b : X' \to X$ such that $H^i(Lb^*E)$ is a perfect $\mathcal{O}_{X'}$-module of tor dimension ≤ 1 for all $i \in \mathbf{Z}$.*

Proof. We strongly urge the reader to find their own proof of the existence of U. Let $\eta \in X$ be the generic point. The restriction of E to η is isomorphic in $D(\kappa(\eta))$ to a finite complex $V^\bullet$ of finite dimensional vector spaces with zero differentials. Set $r_i = \dim_{\kappa(\eta)} V^i$. Then the perfect object E' in $D(\mathcal{O}_X)$ represented by the complex with terms $\mathcal{O}_X^{\oplus r_i}$ and zero differentials becomes isomorphic to E after pulling back to η. Hence by Derived Categories of Schemes, Lemma 0DI8 there is an open neighbourhood U of η such that $E|_U$ and $E'|_U$ are isomorphic. This proves the first assertion. The second follows from the first and Lemma 0ESQ as any nonempty open is scheme theoretically dense in the integral scheme X. □